

Small world theory and the World Wide Web

In today's digital world, people and organizations connect through the World Wide Web, which works like a small-world network. Understanding how these network features influence online interactions is important for improving communication, social media marketing (SMM), and search engine optimization. This is highly relevant, because it helps companies and other organizations use the web to build stronger relationships with customers and markets. ([Ashley and Tuten, 2015](#); [Hoffman and Fodor, 2010](#); [Dominici, 2009](#)).

The existing literature on this topic so far have focused mainly at the tools and methods used to make social media marketing (SMM) ([Evans, 2012](#); [Tuten & Solomon, 2014](#)). Other studies focus on finding better strategies for different types of companies and markets ([Mangold & Faulds, 2009](#)), or on how SMM and the web influence business planning and strategy ([Berthon et al., 2012](#)). Some studies have shown that weak ties in online social networks can have a similar impact as strong ties ([Goldenberg et al., 2001](#); [Henning-Thurau et al., 2004](#)). Other studies focused on whether it is possible to apply small world properties to online social networks while pointing out the key variables of website centrality in that context ([DD Fatta et al, 2016](#)).

Although small-world theory has been applied to online marketing and social media structures, its connection to cybersecurity and misinformation spread remains less studied. While short average path length makes communication efficient, it also increases the frequency of cyberattacks, fake news and other types of online attacks. It would be crucial to understand exactly how do small-world network properties influence the spread of misinformation and cybersecurity risks across online systems?

This research could be used by IT teams and cybersecurity specialists to develop safer online systems. It could also be useful for policymakers, because they could create more effective regulations in order to better protect users if they had a deeper understanding of how false information spreads online. Social media companies could enhance content moderation and strengthen network security. Ultimately, everyday users would also benefit from a safer and more reliable online space with less misinformation and fewer cyber risks.

References:

- [1] Ashley, C., & Tuten, T. (2015). Creative strategies in social media marketing: An exploratory study of branded social content and consumer engagement. *Psychology & marketing*, 32(1), 15-27.
- [2] Berthon, P. R., Pitt, L. F., Plangger, K., & Shapiro, D. (2012). Marketing meets Web 2.0, social media, and creative consumers: Implications for international marketing strategy. *Business horizons*, 55(3), 261-271.
- [3] Dominici, G. (2009). E-marketing. *Analisi dei cambiamenti dai modelli di business al mix operativo*, FrancoAngeli, Milan.
- [4] Evans, D. (2012). *Social media marketing: An hour a day*. John Wiley & Sons.
- [5] Fatta, D. D., Caputo, F., Evangelista, F., & Dominici, G. (2016). Small world theory and the World Wide Web: linking small world properties and website centrality. *International Journal of Markets and Business Systems*, 2(2), 126-140.
- [6] Goldenberg, J., Libai, B., & Muller, E. (2001). Talk of the network: A complex systems look at the underlying process of word-of-mouth. *Marketing letters*, 12(3), 211-223.
- [7] Hennig-Thurau, T., Gwinner, K. P., Walsh, G., & Gremler, D. D. (2004). Electronic word-of-mouth via consumer-opinion platforms: what motivates consumers to articulate themselves on the internet?. *Journal of interactive marketing*, 18(1), 38-52.
- [8] Hoffman, D. L., & Fodor, M. (2010). Can you measure the ROI of your social media marketing?. *MIT Sloan management review*.
- [9] Mangold, W. G., & Faulds, D. J. (2009). Social media: The new hybrid element of the promotion mix. *Business horizons*, 52(4), 357-365.
- [10] Tuten, T.L. and Solomon, M.R. (2014) Social Media Marketing. *Sage, London*.